## Radxa CM4 on DF40C sockets

Pin allocation: docs/head-pcb.md section 4. Unused pins not connected, as on the NANO-A

J9

| | | | |
|---|---|---|---|
| GND | 2 | 1 | GND |
| | Ethernet_Pair1_P | Ethernet_Pair3_P | |
| | Ethernet_Pair1_N | Ethernet_Pair3_N | |
| GND | 6 | 5 | GND |
| | Ethernet_Pair0_N | Ethernet_Pair2_N | |
| | Ethernet_Pair0_P | Ethernet_Pair2_P | |
| GND | 10 | 9 | GND |
| | Ethernet_SYNC_IN(1.8v) | Ethernet_nLED3(3.3v) | |
| | Ethernet_SYNC_OUT(1.8v) | Ethernet_nLED2(3.3v) | |
| | EEPROM_nWP | Ethernet_nLED1(3.3v) | |
| GND | 14 | 13 | GND |
| SAI2_LRCK | GPIO26 | GPIO21 | SAI2_SDO |
| SPI2_MOSI | GPIO19 | GPIO20 | SAI2_SDI |
| SPI_INT | GPIO6 | GPIO16 | |
| GND | 18 | 17 | SPI2_CLK |
| SPI_CS | GPIO5 | GPIO12 | |
| CAM_SDA | ID_SD | ID_SC | CAM_SCL |
| | GPIO11 | GPIO7 | SERVO_PWM |
| | GPIO9 | GPIO8 | |
| GND | 24 | 23 | GND |
| EYE_DATA | GPIO10 | GPIO25 | |
| SPI2_MISO | GPIO22 | GPIO24 | UART7_RX |
| | GPIO27 | GPIO23 | UART7_TX |
| FAN_PWM | GPIO17 | GPIO18 | SAI2_SCLK |
| GND | 30 | 29 | |
| SPI2_CS | GPIO4 | GPIO14 | |
| I2C8_SCL | GPIO3 | GPIO15 | |
| I2C8_SDA | GPIO2 | SD_CLK | |
| GND | 36 | 35 | GND |
| | SD_CMD | SD_DAT3 | |
| | SD_DAT5 | SD_DAT0 | |
| GND | 40 | 39 | GND |
| | SD_DAT4 | SD_DAT1 | |
| | SD_DAT7 | SD_DAT2 | |
| | SD_DAT6 | 43 | GND |
| GND | 44 | SD_VDD_Override | |
| | Reserved | SD_PWR_ON | |
| CM4_3V3 | GPIO_VREF(1.8v/3.3v_Input) | +5v_(Input) | +5V |
| | SCL0 | +5v_(Input) | +5V |
| | SDA0 | +5v_(Input) | +5V |
| CM4_3V3 | +3.3v_(Output) 600mA Max | +5v_(Input) | +5V |
| | +3.3v_(Output) | +5v_(Input) | +5V |
| | +1.8v_(Output) 600mA Max | +5v_(Input) | +5V |
| | +1.8v_(Output) | WiFi_nDisable | |
| CM4_RESET | RUN_PG | BT_nDisable | |
| | AnalogIP1 | nRPIBOOT | |
| | AnalogIP0 | nPI_LED_PWR | |
| GND | 60 | 59 | CAM_GPIO |
| | nEXTRST | Global_EN | |

NB SD signals are only available on modules without eMMC

CM4 socket 1–100

J10

| | | | |
|---|---|---|---|
| | 102 | 101 | |
| | PCIe_CLK_nREQ | USB_OTG_ID | |
| | 104 | 103 | USB_DN |
| | Reserved | USB2_N | |
| | 106 | 105 | USB_DP |
| | Reserved | USB2_P | |
| GND | 108 | 107 | GND |
| | GND | PCIe_nRST | |
| | 110 | 109 | |
| | PCIe_CLK_P | VDAC_COMP | |
| | PCIe_CLK_N | 113 | |
| GND | 114 | 113 | GND |
| | GND | CAM1_D0_N | |
| | 116 | 115 | |
| | PCIe_RX_P | CAM1_D0_P | |
| | PCIe_RX_N | 117 | |
| GND | 118 | 119 | GND |
| | GND | CAM1_D1_N | |
| | 120 | 121 | |
| | PCIe_TX_P | CAM1_D1_P | |
| | PCIe_TX_N | 123 | |
| GND | 124 | 125 | GND |
| | GND | CAM1_C_N | |
| CSI_D0_N | 126 | 127 | |
| | CAM0_D0_N | CAM1_C_P | |
| CSI_D0_P | 128 | 129 | |
| | CAM0_D0_P | 131 | GND |
| GND | 130 | GND | |
| | GND | CAM1_D2_N | |
| CSI_D1_N | 132 | 133 | |
| | CAM0_D1_N | CAM1_D2_P | |
| CSI_D1_P | 134 | 135 | |
| | CAM0_D1_P | 137 | GND |
| GND | 136 | GND | |
| | GND | CAM1_D3_N | |
| CSI_C_N | 138 | 139 | |
| | CAM0_C_N | CAM1_D3_P | |
| CSI_C_P | 140 | 141 | |
| | CAM0_C_P | HDMI1_HOTPLUG | |
| GND | 142 | 143 | |
| | GND | HDMI1_SDA | |
| | 144 | 145 | |
| | HDMI1_TX2_P | HDMI1_SCL | |
| | HDMI1_TX2_N | 147 | |
| GND | 148 | 149 | GND |
| | GND | HDMI1_CEC | |
| | 150 | 151 | |
| | HDMI1_TX1_P | HDMI0_CEC | |
| | HDMI1_TX1_N | 153 | |
| GND | 154 | 155 | GND |
| | GND | DSI0_D0_N | |
| | 156 | 157 | |
| | HDMI1_TX0_P | DSI0_D0_P | |
| | HDMI1_TX0_N | 159 | |
| GND | 160 | 161 | GND |
| | GND | DSI0_D1_N | |
| | 162 | 163 | |
| | HDMI1_CLK_P | DSI0_D1_P | |
| | HDMI1_CLK_N | 165 | |
| GND | 166 | 167 | GND |
| | GND | DSI0_C_N | |
| | 168 | 169 | |
| | HDMI0_TX2_P | DSI0_C_P | |
| | HDMI0_TX2_N | 171 | |
| GND | 172 | 173 | GND |
| | GND | DSI1_D0_N | |
| | 174 | 175 | |
| | HDMI0_TX1_P | DSI1_D0_P | |
| | HDMI0_TX1_N | 177 | |
| GND | 178 | 179 | GND |
| | GND | DSI1_D1_N | |
| | 180 | 181 | |
| | HDMI0_TX0_P | DSI1_D1_P | |
| | HDMI0_TX0_N | 183 | |
| GND | 184 | 185 | GND |
| | GND | DSI1_C_N | |
| | 186 | 187 | |
| | HDMI0_CLK_P | DSI1_C_P | |
| | HDMI0_CLK_N | 189 | |
| GND | 190 | 191 | GND |
| | GND | DSI1_D3_N | |
| | 192 | 193 | |
| | DSI1_D3_N | DSI1_D2_N | |
| | DSI1_D3_P | 195 | |
| GND | 196 | 197 | GND |
| | GND | DSI1_D2_P | |
| | 198 | 199 | |
| | HDMI0_SCL | HDMI0_SDA | |
| | 200 | | |

High Speed Serial

CM4 socket 101–200

GPIO_VREF (78) on the module's own 3.3 V output (84; 86 is the same rail, left open): 3.3 V GPIO, as the NANO-A's H2 jumper

H5

Radxa CM4

## Head bundle in, 5 V buck

LMR51635Y, 1.1 MHz, 4.7 uH (Ti table 8-2: 3.3 uH), 82k/15k sets 5.17 V, EN tied to VIN. F6 (2 A) on the bridge fuses this feed. JP1 joins the buck to the 5 V rail: cut it to feed 5 V from outside (J12). The feedback divider stays on the buck side

J1

VBAT_IN 1, GND 2, LINK_RX 3, LINK_TX 4 — Head bundle 1 +BATT, 2 GND, 3 bridge TX, 4 bridge RX (bridge J6 order)

U2 LMR51635YDDCR: VIN, EN, CB, SW, FB, GND

BUCK_CB — C5 100n 50V — BUCK_SW

BUCK_SW — L1 4.7u — 5V_BUCK

5V_BUCK — R1 82k — BUCK_FB

BUCK_FB — R2 15k — GND

5V_BUCK — C6 47u 10V — GND; C7 47u 10V — GND; C8 47u 10V — GND

D1 SMBJ30A: VBAT_IN — GND

C1 10u 50V X7R; C2 10u 50V X7R; C17 47u 50V; C4 100n 50V — VBAT_IN — GND

JP1: 5V_BUCK — +5V, cut for ext 5V

PWR_FLAG — VBAT_IN; PWR_FLAG — GND; PWR_FLAG — +5V

## Fan and eye LED

Fan: low side, PWM1_CH2 (pin 50). Eye: WS2812-type, SPI1 MOSI (pin 44), 3.3 V data

D2 B5819W: +5V — FAN_N

J6: +5V, FAN_N — Fan 1 +5V, 2 FAN−

FAN_PWM — R5 100R — FAN_G

Q1 AO3400A: FAN_G 1, FAN_N 3, 2 GND

FAN_G — R6 100k — GND

J4: +5V, EYE_DATA, EYE_DIN, GND — Eye LED 1 +5V, 2 DATA, 3 GND

EYE_DATA — R7 330R — EYE_DIN

## I2C8: head IMU module, ToF, breakout

Generic BMI088 module on short header pins, front side (rev A position): its own pull-ups and PS strap (set to IIC), SDO to GND: accelerometer 0x18, gyroscope 0x68 (the module joins SDO1 and SDO2). No bus pull-ups on the board: the module and the ToF breakout carry them. J3: spare I2C8 on 2.0 mm pin holes (PH pitch), Qwiic order

U5 BMI088 module: +3V3, GND, I2C8_SCL, I2C8_SDA, GND; VCC, GND, SCK, SDA, SDO, csA, csG; INT4, INT3, INT2, INT1

J2: GND, +3V3, I2C8_SDA, I2C8_SCL — ToF VL53L8CX breakout, Qwiic order

J3: GND, +3V3, I2C8_SDA, I2C8_SCL — I2C pins Spare I2C8, same order; pin holes, no connector

## Power LED, mounting holes

M2 holes, one per corner, not connected

+5V — R12 1k — LED_A; GND — D3 green — LED_A

H1 M2; H2 M2; H3 M2; H4 M2

## SPI breakout

SPI2 M1 for a future SPI device, untested. Wire pads on the bottom, under the module, along the top edge. CLK GPIO12 (CM4 31), MOSI GPIO13 (28), MISO GPIO22 (46), CS0 GPIO4 (54), INT GPIO6 (30), spare GPIO5 (34)

J13: GND, +5V, 3V3, SPI2_MOSI, SPI2_MISO, SPI2_CLK, SPI2_CS0, SPI_INT, SPI_IO — SPI pads 1 GND, 2 5V, 3 3V3, 4 MOSI, 5 MISO, 6 CLK, 7 CS0, 8 INT, 9 spare GPIO

## 3.3 V

AMS1117-3.3 from 5 V, as on the NANO-A. Camera, ToF, IMU, mic, I2C breakout

U3 AMS1117-3.3: +5V 3, VI; GND 1; VO 2 — +3V3

C9 10u 25V: +5V — GND; C10 22u 25V: +3V3 — GND; C11 100n 50V: +3V3 — GND

## Speaker amplifier

MAX98357A on SAI2. SD_MODE 1M to 5 V: (L+R)/2 mono. GAIN open = 9 dB: fit R4 (100k to GND) for 15 dB. JP3 feeds it: cut JP3 to use an external amplifier module on J11

JP3: +5V — AMP_5V. cut for ext amp

U4 MAX98357AETE+T: SAI2_SDO 1, SAI2_SCLK 16, SAI2_LRCK 14, AMP_SD 4, AMP_GAIN 2; DIN, BCLK, LRCLK, SD_MODE, GAIN_SLOT, GND 11, PAD 17, VDD, OUTP 9, OUTN 10 — SPK_P, SPK_N

C12 10u 25V: AMP_5V — GND; C13 100n 50V: AMP_5V — GND

R3 1M: AMP_5V — AMP_SD

R4 100k: AMP_GAIN — GND

J5: SPK_P 1, SPK_N 2 — Speaker. Bridge-tied output: neither lead to GND

PWR_FLAG — AMP_5V

## Microphone

INMP441 on SAI2. L/R tied to GND on the mic side (left slot)

J7: +3V3 1, GND 2, SAI2_SCK 3, SAI2_LRCK 4, SAI2_SDI 5 — Microphone 1 3V3, 2 GND, 3 SCK, 4 WS, 5 SD

## Camera FPC

Pi 15-pin camera order. CAM0 lanes (Radxa CSI1). I2C6 with 4.7k pull-ups, CAM_GPIO enable. 100 ohm pairs

J8: GND 1, CSI_D0_N 2, CSI_D0_P 3, GND 4, CSI_D1_N 5, CSI_D1_P 6, GND 7, CSI_C_N 8, CSI_C_P 9, GND 10, CAM_GPIO 11, 12, CAM_SCL 13, CAM_SDA 14, +3V3 15 — MIPI camera. GND

R10 4.7k: +3V3 — CAM_SCL

R11 4.7k: +3V3 — CAM_SDA

C14 100n 50V: +3V3 — GND

Pin 12 (camera clock) not connected: the module has its own oscillator

## External fallbacks

Wire pads, 2.54 mm. J12: cut JP1 and feed 5 V from an external step-down. J11: cut JP3 and wire an amplifier module, pins in the generic module's order; J11 is bottom-side solder pads, between the IMU module's pin rows

J12: VBAT_IN 1, GND 2, +5V 3, GND 4 — Ext 5V. J12: 1 B+ out, 2 GND, 3 5V in, 4 GND

J11: SAI2_LRCK 1, SAI2_SCLK 2, SAI2_SDO 3, 4, 5, GND 6, +5V 7 — Ext amp. J11: LRC, BCLK, DIN, GAIN (open), SD (open: module's own pull-up), GND, Vin

## Link series resistors, test pads

Series R in each UART7 line: 1k on RX limits current into an unpowered CM4 from the bridge, 330R on TX. TP1/TP2 (bottom): short to TP3 to reset; hold TP2 low through a reset for maskrom. TP4–TP6 (bottom): USB D−, D+, GND of the download port, for rkdeveloptool/m maskrom. TP7: GPIO7 / PWM0_CH0_M1 (pin 37) for a servo

R13 1k: UART7_RX — LINK_RX; R14 330R: UART7_TX — LINK_TX

TP1 USB_DN; TP2 USB_DP; TP3 GND; TP4 GND; TP5 GND; TP6 GND; TP7 SERVO_PWM

Bootstrapped by scripts/kicad/gen_head_sch.py; edit this file from now on

Design note: docs/head-pcb.md

File: head_ext.kicad_sch

**Title: Dukki head carrier, external IMU**

Size: A2 | Date: 16/10/2026 | **Rev: B**

KiCad E.D.A. 10.0.1 | Id: 1/1